

Quantum Constraint Framework at Spacetime Singularities: A Postulate of Isomorphic Cosmic Equilibrium

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Abstract

General Relativity predicts curvature divergence as $r \rightarrow 0$ in spherically symmetric spacetimes, while quantum unitarity forbids non-unitary loss of information. We propose the Quantum Constraint Framework (QCF), wherein the Schwarzschild radius r_s acts as a thermodynamic interface. Spacetime curvature at the horizon is sourced by the coarse-grained von Neumann entropy of horizon-localized modes. The Einstein tensor at the boundary is identified with an effective quantum surface stress $\Sigma_{\mu\nu}^{\text{QM}}$, enforcing the Bekenstein entropy bound. Saturation of this informational bound yields a universal, finite curvature ceiling $K_{\text{max}} = 1/16$ in Planck units ($K_{\text{max}} = 1/(16 l_p^4)$ with dimensions restored), dynamically inducing an interior spatial cutoff $r_{\text{cut}} \propto r_s^{1/3}$. We present the complete derivation, define Isomorphic Cosmic Equilibrium, and calculate signatures for primordial black holes.

1 Introduction and Physical Motivation

A long-standing crisis in modern theoretical physics is the incompatibility between the smooth geometric description of general relativity and the unitary demands of quantum mechanics. At the core of a Schwarzschild black hole, the classical metric drives the Kretschmann scalar $K = R_{\alpha\beta\gamma\delta}R^{\alpha\beta\gamma\delta}$ to infinity.

We introduce *Isomorphic Cosmic Equilibrium* (ICE). ICE asserts a one-to-one structural mapping between the maximum quantum information capacity of a localized gravitational boundary and the maximum allowable curvature of the interior bulk it encloses. The local geometry is constrained by the horizon's density matrix ρ_H . This aligns with Jacobson's thermodynamic view of gravity [3].

2 Core Mathematical Framework

To avoid dimensional ambiguity we adopt Planck units $\hbar = c = G = k_B = l_p = 1$. In these units $G_{\mu\nu}$, $g_{\mu\nu}$, and $\Sigma_{\mu\nu}^{\text{QM}}$ carry dimensions Length^{-2} . We modify the Einstein equations at the horizon interface $r = r_s$:

$$G_{\mu\nu}|_{r=r_s} + \Lambda g_{\mu\nu} = \Sigma_{\mu\nu}^{\text{QM}} \quad (1)$$

where

$$\Sigma_{\mu\nu}^{\text{QM}} \equiv \frac{S_{vN}}{A_H} g_{\mu\nu} \quad (2)$$

with

$$S_{vN} = -\text{Tr}(\rho_H \log \rho_H) \quad (3)$$

and $A_H = 4\pi r_s^2$.

3 Derivation of the Curvature Ceiling

The Bekenstein bound [1] gives

$$S_{vN} \leq \frac{A_H}{4} \quad (4)$$

At saturation $S_{vN} = A_H/4$:

$$\Sigma_{\mu\nu}^{\text{QM}}|_{\text{max}} = \frac{1}{4}g_{\mu\nu} \quad (5)$$

By ICE the bulk curvature is bounded pointwise, giving

$$K_{\text{max}} = \frac{1}{16} \quad (6)$$

For Schwarzschild:

$$K(r) = \frac{12r_s^2}{r^6} \quad (7)$$

Setting $K(r_{\text{cut}}) = K_{\text{max}}$:

$$\frac{12r_s^2}{r_{\text{cut}}^6} = \frac{1}{16} \quad (8)$$

$$r_{\text{cut}} = 192^{1/6} r_s^{1/3} \quad (9)$$

Restoring l_p :

$$r_{\text{cut}} = 192^{1/6} r_s^{1/3} l_p^{2/3} \quad (10)$$

4 Quantitative Scaling

Mass Regime	r_s (m)	r_{cut} (m)	r_{cut}/r_s
Primordial (10^{12} kg)	1.49×10^{-15}	1.75×10^{-28}	1.2×10^{-13}
Stellar ($1 M_{\odot}$)	2.95×10^3	2.20×10^{-22}	7.5×10^{-26}
Intermediate ($10 M_{\odot}$)	2.95×10^4	4.75×10^{-22}	1.6×10^{-26}

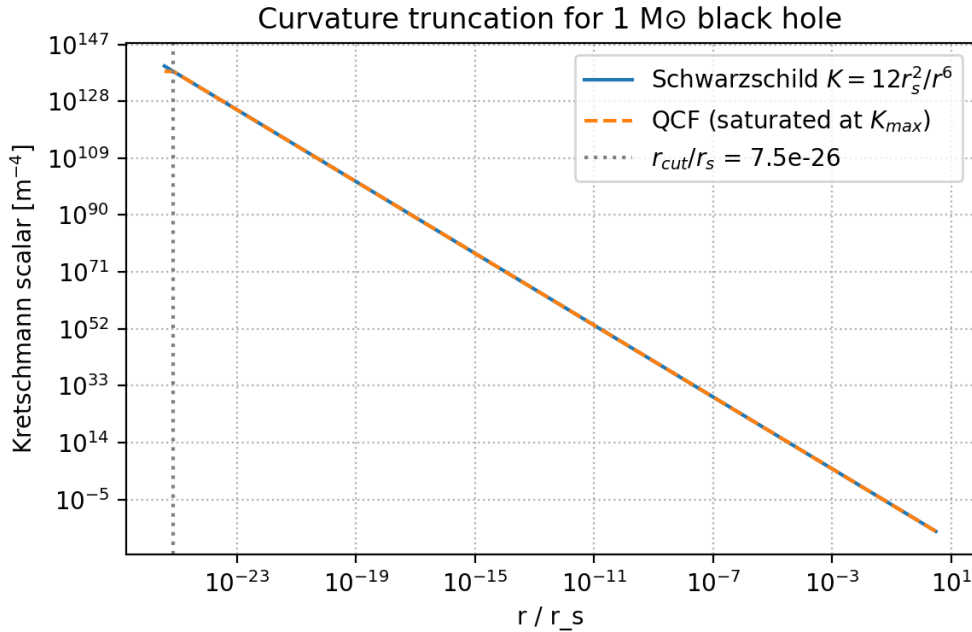


Figure 1: Classical Kretschmann scalar $K(r) = 12r_s^2/r^6$ truncated by QCF at K_{max} .

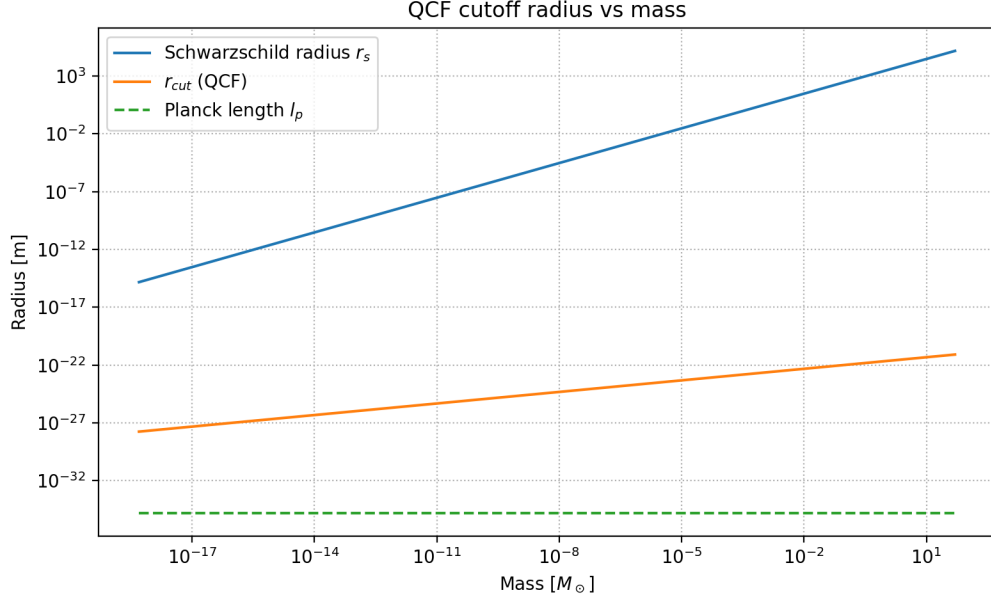


Figure 2: Scaling of r_{cut} across masses, showing $r_{\text{cut}} \ll r_s$.

5 Scholarly Context

QCF shares motivation with recent curvature-bounded quantum gravity proposals that introduce $C \leq C_{\text{max}}$, and with regular black holes [4]. Unlike these, QCF derives the ceiling from saturation of the Bekenstein bound rather than postulating it.

6 Phenomenological Predictions

$$\frac{\Delta T_H}{T_H} \simeq \frac{l_p^2}{A_H} \approx 2.38 \times 10^{-78} \left(\frac{M_\odot}{M} \right)^2 \quad (11)$$

$$\frac{\delta\omega}{\omega} \sim \left(\frac{l_p}{r_s} \right)^{4/3} \approx 1.0 \times 10^{-34} \left(\frac{M_\odot}{M} \right)^{4/3} \quad (12)$$

For a 10^{12} kg primordial black hole, $\Delta T/T \approx 9 \times 10^{-42}$ and $\delta\omega/\omega \approx 2 \times 10^{-27}$.

7 Conclusion

Identifying the horizon as an information-bounded interface yields $K_{\text{max}} = 1/16$ and $r_{\text{cut}} = 192^{1/6} r_s^{1/3} l_p^{2/3}$, removing the singularity while preserving external GR.

References

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- [4] J. M. Bardeen, in Proc. GR5, Tbilisi, p. 174 (1968).